CURRICULUM VITAE

YUANJING ZHAO

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PROFILE

My research interests centre on sustainable energy systems, CO2 capture and utilisation, and process systems engineering, particularly electrochemical CO2 conversion and renewable methanol production. Current work focuses on surrogate-based optimisation, process digitalisation and multi-objective optimisation of complex multi-physics systems. Memberships: IChemE Student Member (No. 200290733); CIESC Member (No. E350023430M).

EDUCATION

PhD, Process Systems Engineering
09/2023 – 07/2026
University of Surrey — England

Visiting Student, Department of Chemical Engineering
05/2024 – 09/2024
University of Cagliari — Italy

MSC, Advanced Chemical Engineering
08/2020 – 12/2021
Imperial College London — England

Research Assistant, Applied Sciences
06/2018 – 12/2018
RWTH Aachen University | Jülich Supercomputing Centre — Germany

BEng, Chemical Engineering and Technology
09/2016 – 06/2020
Sun Yat-sen University, GPA: 3.8/4 — China

SKILLS

Python · MATLAB · GAMS · COMSOL · gPROMS · Aspen

AWARDS

Fully Funded Doctoral Studentship, *University of Surrey*	2023-2027
Best Oral Presentation Award, *SCN Symposium*	2026
Funding Award, *Turing Scheme*	2024
Best Oral Presentation Award, *31st CSCST Annual Conference*	2024
Certification of Fundamentals of Deep Learning, *NVIDIA*	2023
Outstanding Graduation Design, *Sun Yat-sen University*	2020
National First Prize, *China Undergraduate Mathematical Contest in Modeling (CUMCM)*	2019
The 1st Prize, *Challenge Cup of National Undergraduate Science and Technology*	2019
First Class Scholarship, *Sun Yat-sen University*	2017 & 2018
The 1st Prize, *MathorCup Mathematical Modelling Competition*	2018
AICHE Chem-E-Car of the Year 2018, *7th China mainland*	2018

LANGUAGES

English ●●●●◐
Cantonese ●●●●●
Mandarin ●●●●●

RESEARCH PROJECTS

Integrated Green Methanol Process Modelling and Assessment
09/2023-09/2027
University of Surrey — Guildford, England
Evidence link >

- Developed and validated an integrated Aspen Plus–MATLAB framework to compare two electrochemical and one conventional methanol production route, incorporating symbolic-regression models for CO_2 and water electrolysers.

Mechanistic Modelling and Scale-Up of CO_2 Electrolysers
06/2024-07/2027
Process Systems Enterprise, SIEMENS — London, England

- Collaborated with Siemens on the mechanistic development and gPROMS modelling of multiple CO2 electrolyser technologies.
- Developed scale-up methodologies and integrated electrolyser models with chemical production systems to evaluate system-level performance and support industrial deployment.

PINN–COMSOL Integration Project
04/2025–07/2026
University of Surrey — Guildford, England

- Lead the restoration, consolidation and version control of a PINN codebase for physics-informed modelling and planned COMSOL integration.
- Establish reproducible training, validation and supervised-data workflows, supported by versioned model experiments and structured simulation reporting.

Advanced Process Operations Research
12/2021-12/2022
Imperial College London — London, England

- Established a time-dependent dynamic model for a photobioreactor and conducted multi-objective optimisation analysis to maximise the yield of fatty acid methyl ester and microalgae biomass.
- The results showed that the average confidence of the parameters in the green algae cultivation model was increased by about 20%.

WORK EXPERIENCE

Chemical Engineer / Product Manager
12/2021 – 08/2023
Research Institute of Tsinghua University — Guangzhou, China

- Developed an improved algorithm (CNLP) to solve bilinear optimisation problems in petrochemical planning and scheduling.
- Developed a time-dependent method for the integrated scheduling optimisation of event-driven continuous gasoline and crude-oil blending processes.
- Led the functional and interface design of an industrial optimisation platform, packaged optimisation algorithms for deployment, coordinated back-end integration and maintenance with IT developers, and managed the supporting database.

Teaching Assistant
2023-2026
University of Surrey — Guildford, England

- Supported teaching in chemical reaction engineering, process modelling and simulation, engineering systems, mathematics and computing across seven undergraduate and postgraduate modules.

PUBLICATIONS

1. Zhao, Y., et al. Electrified carbon-neutral methanol production pathways: A multiscale techno-economic-environmental assessment based on the UK energy system. Advanced Chemical Engineering (2026), e70009.
2. Zhao, Y., et al. From thermal to electrochemical CO_2-to-methanol conversion: A comprehensive review of process technologies, techno-economics, and life-cycle performance. Carbon Capture Science & Technology (2026), 100603.
3. Zhao, Y., et al. Multiscale modelling and techno-economic analysis of green methanol synthesis from CO_2 and renewable hydrogen. ESCAPE-35, Ghent, Belgium (2025), 247–248.
4. Zhao, Y., et al. Interpretable surrogate-assisted multi-objective optimisation framework for electrolyser-integrated green methanol process design. *Energy & Environmental Science*, under review (2026).
5. Zhao, Y., et al. A Shared–Private Physics-Informed DeepONet with Hard Electroneutrality and Interface Constraints for CO_2 Electrolyser Modelling. *Joule*, under review (2026).
6. Tian, X., Zhao, Y., et al. 3D Mechanical-Electrochemical Modelling and Optimisation of Anisotropic Transport-Reaction Processes in Electrochemical CO_2 Reduction Electrolysers. Joule, (under review).
7. Wang, S., Peng, Y., Zhao, Y., et al. From Process Simulation to Circular Value: Environmental-Economic Assessment of Advanced Recycling for End-of-Life Photovoltaics. Sustain. Prod. Consum. 2026. DOI: 10.1016/j.spc.2026.06.004.
8. Zhu, P., Huang, Y., Wang, S., Zhao, Y., et al. Multi-Objective Topology Optimization of Solid Oxide Fuel Cell Flow Channels for Superior Performance. Chem. Eng. J. 2026. DOI:10.1016/j.cej.2026.177814.
9. Yang, N., Hu, Y., Zhao, Y., et al. Sustainable carbon-based composites for hydrogen evolution: Catalyst engineering and performance enhancement pathways. Renewable Energy 262 (2026), 125425.
10. Peng, Z., Peng, J., Zhao, Y., et al. Enhancing the technical and economic performance of proton exchange membrane fuel cells through three critical advancements. Systems and Control Transactions 4 (2025), 352–357.